Design Equations and Parametric Sizing for Hierarchical Coordination in Large Autonomous Space Swarms

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***Abstract*—Scaling autonomous coordination beyond 10^4 spacecraft requires sizing relationships that current literature does not provide. We derive closed-form message-layer equations for hierarchical swarm coordination (10^3–10^5 nodes) and distinguish two feasibility layers—message-layer byte budget (η) and physical-layer schedulability (γ, TDMA airtime)—with an explicit test for half-duplex airtime constraints. The framework decomposes overhead into topology-dependent (η_0) and workload-dependent (η_{cmd}) components; coordinator ingress is sized as $C_{\text{coord}} \geq k_c S_{\text{eph}} \times 8/(T_c\gamma)$; and correlated loss recovery is characterized via a Gilbert-Elliott Markov chain. Under a representative instantiation ($k_c = 100$, $S_{\text{eph}} = 256$ B, $T_c = 10$ s, centralized command generation): $\eta_0 \approx 5\%$; stress-case $\eta_S \approx 46\%$ at continuous duty ($d = 1$, episodic <1% of operational time); routine operations ($d = 0.01$–0.10: station-keeping, orbit-raising) yield $\eta \approx 5$–10%; coordinator ingress ≈ 27 kbps at standards-derived $\gamma = 0.76$; GE inter-cycle P95 recovery in 4 cycles (with illustrative GE parameters pending ISL-specific measurement); AoI P99 $= 440$ s under exception telemetry. CCSDS Proximity-1 standards-grounded derivation yields $\gamma = 0.76$ (24 kbps), confirming 30 kbps as the minimum viable coordinator PHY rate; a link budget justifies the 1 kbps RF-backup design point from UHF omnidirectional parameters.**

***Index Terms*—Space systems coordination, swarm robotics, hierarchical control, design equations, parametric sizing, mega-constellations, autonomous spacecraft**

I. Introduction

A. The Coordination Scaling Problem

Coordinating large autonomous spacecraft fleets is a significant open challenge. The largest constellation, SpaceX Starlink (∼7,000 satellites, mid-2024), uses centralized ground coordination [1], but already encounters conjunction challenges [2]. Expansions to 42,000+ satellites [3], [4], and proposed infrastructure concepts at 10^5–10^6 units [5], [6], create qualitatively different coordination problems where latency, bandwidth, and failure recovery exhibit nonlinear scaling [7].

To our knowledge, no prior work provides closed-form parametric sizing relationships for coordination architectures across 10^3–10^5 nodes with byte-level traffic accounting under a fixed per-node budget. Swarm robotics studies tens to hundreds of agents [8], [10]; constellation management addresses ∼10,000 nodes [9]; networking literature treats mega-constellation routing [29], [37], [38]—but the combination of autonomous coordination, byte-level accounting, and 10^4–10^5 scale is underexplored.

Individual author names and affiliations will be provided for final publication per IEEE policy.

TABLE I
Key Notation

Symbol	Definition
N	Fleet size (nodes)
k_c	Cluster size (nodes per coordinator)
T_c	Coordination cycle period (10 s default)
C_{node}	Per-node bandwidth allocation (1 kbps default)
S_{eph}	Status report size (256 B)
C_{coord}	Coordinator ingress link rate (kbps)
γ	MAC efficiency (Eq. 14; 0.76 at 24 kbps, §IV-I)
d	Campaign duty factor ($d \in [0, 1]$; fraction of active-command cycles)
η	Protocol overhead ($= \eta_0 + \eta_{\text{cmd}}$, beyond baseline)
η_{total}	Total utilization ($= \eta + 20.5\%$ baseline)
p_{exc}	Exception reporting probability per node per cycle
M_r	Maximum retransmission attempts per cycle
p_{BG}, p_{GB}	GE state transition probabilities
p_G, p_B	Per-packet loss probability in Good/Bad state
α_{RX}	Ingress fraction of T_c (0.918 at $k_c = 100$)
q	Unicast fraction of commands (Eq. 8)
L_{cmd}	Command stagger cycles (Eq. 7)
f_{RF}	Fraction of fleet in RF-backup mode
F	Number of orthogonal RF channels
R	Spatial reuse factor

B. Research Questions

This study addresses three interrelated research questions centered on characterizing hierarchical coordination scaling:

RQ1: What are the scaling properties of hierarchical coordination architectures for autonomous space swarms across the 10^3 to 10^5 node range, as characterized by protocol overhead, latency distribution, and fault tolerance?

RQ2: What cluster size and coordinator duty cycle parameters are favorable for hierarchical coordination under the assumed message model, and how do they interact?

RQ3: Where does hierarchical protocol overhead fall relative to reference baselines representing centralized ground processing and global-state mesh dissemination, and what coordinator link capacity is required to sustain hierarchical coordination at scale?

C. Contributions

This paper derives a two-layer feasibility framework (byte budget + airtime schedulability) with closed-form sizing equations that accept arbitrary workload semantics, command models, and MAC parameters. The framework decomposes overhead into topology-dependent (η_0) and workload-dependent (η_{cmd}) components, sizes coordinator ingress from cluster

TABLE II
DESIGN EQUATION SCALING ACROSS BANDWIDTH REGIMES (ANALYTICAL)

Metric	1 kbps	10 kbps	100 kbps
η_{stress}	46%	4.6%	0.46%
$\eta_{\text{total,stress}}$	67%	7.1%	0.66%
Coord. bottleneck?	Yes (21 kbps)	No	No
TDMA required?	Yes ($\gamma \geq 0.67$)	No	No
AoI P99 (exception)	440 s	440 s[a]	440 s[a]
GE recovery P95	4 cycles	4 cycles	4 cycles

[a]AoI depends on p_{exc} and T_c, not C_{node}; unchanged across regimes.

parameters, and characterizes correlated loss recovery. Under the framework: architecture-specific overhead ($\eta_0 \approx 5\%$: heartbeats, summaries, election) is small; command traffic dominates the stress-case (>60%) and is topology-invariant *given the assumed workload semantics* (centralized command generation; volume and addressing may differ under alternative decision architectures). A campaign duty factor d parameterizes how frequently commands are active: the stress-case ($d = 1$) bounds continuous operations; realistic campaigns ($d = 0.01$–0.10) reduce overhead to $\eta \approx 5$–10%. Notation: η_0 = architecture-specific overhead; $\eta = \eta_0 + \eta_{\text{cmd}}$ = total protocol overhead beyond baseline telemetry (20.5%); all values are message-layer (MAC overhead scales by $1/\gamma$). The centralized and global-state mesh are intentional bounds. Table II summarizes key metrics across bandwidth regimes: At ≥10 kbps both feasibility layers are non-binding: $\eta_{\text{total}} < 7.1\%$ and CSMA suffices. Unmodeled constraints (antenna scheduling, interference, processing latency at $k_c > 200$) may then dominate. The 1 kbps RF-backup channel (<1% of operational lifetime, but critical for safe-mode) is the *only* regime requiring the TDMA schedulability analysis developed here.

1) **Two-layer feasibility framework** (Sections IV-A–IV-E): sizing equations from message-layer byte budget (Layer 1: η, η_{total}) through half-duplex TDMA airtime schedulability (Layer 2) to coordinator capacity. Architecture-specific overhead $\eta_0 \approx 5\%$ (summaries, heartbeats, election); workload-dependent η_{cmd} yields $\eta_E \approx 6\%$ (routine) to $\eta_S \approx 46\%$ (stress bound). Under centralized command generation (the baseline assumption); cluster-local planning would reduce η_{cmd} but introduce intra-cluster consensus overhead;
2) **Coordinator ingress sizing** (Section IV-A): zero-drop ingress at 21–50 kbps ($\times 1/\gamma$); scheduling disciplines converge to ∼21–25 kbps;
3) **Correlated loss recovery** (Section IV-C): GE intra-cycle recovery 27% (vs. 87.5% i.i.d.); inter-cycle P95 recovery in 4 cycles at $p_{BG} = 0.50$ (mean 1.7); sensitivity sweep over p_{BG} and p_B provides design curves for arbitrary channel burstiness (Fig. 8).

II. RELATED WORK

A. Constellation Coordination

Current constellations (<15,000 nodes) use centralized ground coordination [1], [2], [4]; none has published scalability evidence beyond this regime. ISL routing [21], [29], [38], [46], DTN [36], [39], and Contact Graph Routing [49] address mega-constellation networking. NASA DSA [48] demonstrates onboard decision-making; DARPA F6 [32] and Golkar [33] explored fractionated architectures. Our message sizes align with CCSDS SPP [47].

B. Swarm Robotics, Multi-Agent Theory, and Operational Programs

Network calculus [54] provides worst-case deterministic bounds for periodic traffic; our sizing equations complement this by characterizing the design envelope under stochastic workloads and correlated losses. The GE channel parameterization follows the Lutz et al. two-state Markov framework [52], grounded in ITU-R P.681 shadowing statistics [53].

Most swarm demonstrations involve 10–100 agents [8]; the scalability gap is well documented [11]. LEACH [50] analyzes cluster-head rotation; our coordinator model draws on similar principles under space ISL constraints. Bio-inspired optimization [12]–[14] lacks convergence guarantees at 10^6 scale. Lamport clocks [26] and Raft [27] inform coordinator election; consensus under switching topologies [34], [35], GNN controllers [15], [18], and mean-field games [16], [17] address theoretical scaling. DARPA OFFSET [19], Blackjack [30], and Replicator [20] have demonstrated up to ∼250 units; SWIM [31] provides applicable failure detection.

III. SIMULATION FRAMEWORK

A. Cycle-Aggregated Discrete Event Simulation

We implement a cycle-aggregated simulation in which autonomous nodes exchange messages according to one of four coordination topologies. The simulation advances in coordination cycle increments ($T_c = 10$ s); within each cycle, all N nodes generate messages, coordinator queues are serviced, byte counts are accumulated, and latencies are recorded. The atomic unit is a *message-layer event* (generation, routing, delivery/drop, byte accounting)—not per-packet or per-bit. This enables fast runtimes (∼7 s at $N = 10^5$). Each run models one year of operations.

Verification taxonomy (cf. IEEE 1012 [51]). (1) *Code verification*: DES matches closed-form (<0.1%, Table XIV); DES additionally provides distributional analysis (Fig. 10). (2) *Model verification*: slot-level TDMA simulator confirms superframe budget and loss–capacity interaction (Tables VII, XI). (3) *Standards-grounded* γ: packet-level simulator derives γ from CCSDS framing (Section IV-I). (4) *Remaining gap*: MAC contention and antenna scheduling require NS-3 (Section V-A). All tools are open-source.

Events: status (256 B), commands (512 B), handoffs, exponential failures (2%/yr), collision alerts (128 B). MAC scheduling abstracted via γ (§IV-A). **DES cycle update** (each T_c): generate messages → route → GE state per link → deliver/lose (p_G/p_B), retry $\leq M_r$ → coordinator ingress (fluid server, drop-tail) → update counters.

Tool scope disambiguation. The DES and slot-level TDMA simulator answer *different questions* and should not be conflated. The DES operates at message-layer granularity: it

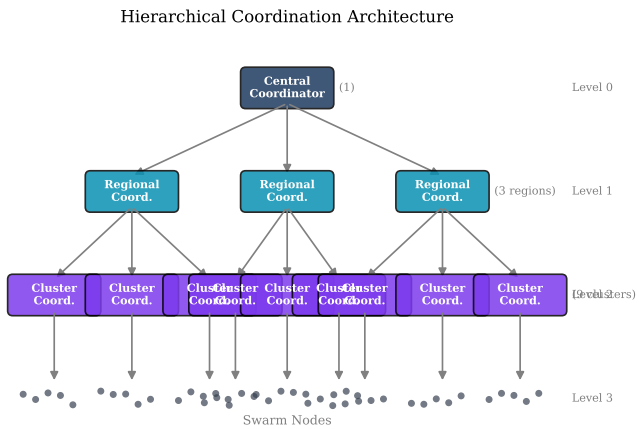

Fig. 1. Four-level hierarchical architecture: Ground → Regional (10–100) → Cluster (10–100) → Node (50–100). Labels: aggregation ratios.

computes η, tracks AoI, and provides distributional analysis of coordinator ingress under stochastic campaigns—but it has no slot structure, no guard intervals, and no half-duplex partitioning. DES-reported "drops" reflect message-layer byte-budget overflow at C_{coord}, not TDMA deadline misses. The slot-level simulator operates at individual TDMA slot granularity: it enforces superframe timing, guard intervals, and half-duplex TX/RX partitioning (Table VII), revealing interactions (e.g., ARQ × TDMA coupling at 24 kbps) that the fluid-server DES cannot capture. The two tools are complementary: DES for mean-value sizing and parametric sweeps; slot-sim for schedulability and worst-case timing.

B. Topology Models

1) Centralized Ground Processing (Reference Baseline): A single ground coordinator modeled as an $M/D/1$ queue [22] with utilization $\rho = Nr/\mu_s$; with c parallel servers, $N_{\max} = c\mu_s/r$ (10^4 at $c = 1$, 10^6 at $c = 100$). This is an intentional compute bound; the binding constraint for centralized architectures is propagation latency and uplink spectrum, not processing. Complexity: $O(N)$.

2) Hierarchical Topology: The hierarchical model implements a four-level tree structure (Fig. 1):

$$\text{Levels: Ground} \rightarrow \text{Regional} \rightarrow \text{Cluster} \rightarrow \text{Node} \tag{1}$$

with configurable fan-out. **Topology dynamics.** The simulation assumes *static* cluster membership for the 1-year duration. This is exact for co-planar formations; for cross-plane configurations, cluster re-association adds <0.5% overhead (quantified in Section V-B). Each cluster coordinator sends a single 512-byte summary per cycle (vs. forwarding k_c individual reports); regional coordinators forward 1024-byte region summaries. The total message count is:

$$M_{\text{total}} = N + \frac{N}{k_c} + \frac{N}{k_c \cdot k_r} \tag{2}$$

where k_c is the cluster size and k_r is clusters per region. The DES models full bidirectional traffic (commands 512 B, heartbeats 64 B, alerts 128 B); the reported $\eta \approx 46\%$ includes both directions. Complexity is $O(N)$.

Coordinator service discipline. Reports processed at $s_{\text{proc}} = 5$ ms/msg ($\mu_c = 200$ msg/s); $D[k_c]/D/1$ batch latency $\leq$ 500 ms. Coordinator rotation: state transfer (10–50 MB) over optical ISL, excluded from η. **Nominal handoff:** 3–5 s (Raft election [27]; quorum = 51). **RF-backup handoff:** Raft election (∼113 s at Slotted ALOHA) + seed handoff (16 s) + state rebuild (1–3 cycles). Total: ∼160 s.

Coordinator failure transient. Optical ISL: 3–5 s election, ≤1 cycle missed. RF-backup: ∼160 s (∼16 cycles missed); triple fault (+ GE bad-state): 1.8×10^{-5}/yr, ∼310 s. AoI impact: +100–200 s, modest vs. P99 = 441 s.

Operational impact of coordinator failure during RF-backup. Members detect coordinator absence within $2T_c$ (20 s) via heartbeat timeout, entering safe hold ($\eta_{\text{total}} \approx 5.7\%$, inertial coasting, broadcast collision alerts). Worst-case delay: 380 s ($160 + 22 \times T_c$). Compound probability of simultaneous ISL outage and coordinator failure: $\sim 6.3 \times 10^{-12}$ s^{-1} (one per 5,000 yr/cluster).

3) Global-State Mesh (Upper Bound on Decentralized Overhead): The global-state mesh (each node maintains full fleet state) requires $O(N^2)$ messages for single-cycle convergence via aggressive gossip [23]—an intentional worst-case upper bound. At $N = 10^5$: ∼73 MB/node/cycle (exceeds the 1 kbps budget). Hierarchical aggregation avoids this via $O(k_c)$ per-node cost.

4) Sectorized Mesh (Local-Neighborhood Monitoring Reference): A sectorized mesh restricts gossip to $k_s = \lceil \sqrt{N} \rceil$ orbital neighbors per sector with capped fanout (cap = 10): $\eta_{\text{sector}} \approx 65$–$67\%$ (stress-case, 3.2% sector coverage). This provides local-neighbor monitoring only—functionally distinct from hierarchical cluster-wide coordination (Table XVI).

C. Node Model

Each node has: 5 W baseline power (15–20 W coordinator mode), 1 kbps coordination bandwidth, exponential failures at 2%/year (MTTF = 50 yr, consistent with LEO small-satellite data [24]), and a 6-element state vector (position, velocity, health, power, cluster). Failed nodes cease communication; the coordinator detects failure within one cycle.

D. Monte Carlo Configuration

30 MC replications per configuration; swept parameters in Table III. Full-participation (no sampling); runtime ∼0.2–7 s per run. Results: means with 95% bootstrap CIs (SD < 0.001% for overhead; tail metrics use the full 30-replication ensemble).

Static topology justification. Fixed cluster membership is acceptable for bandwidth sizing (η depends on k_c and message sizes, not node identities). Re-association overhead: <0.5% (Section V-B). Transient AoI during churn is future work.

The simulation is vectorized (∼7 s at $N = 10^5$). **Not modeled** (captured by γ or identified as future work): MAC contention, link acquisition/pointing, Doppler, orbital perturbations, correlated failures, priority queueing, DTN routing.

TABLE III
SIMULATION PARAMETERS

Parameter	Value	Notes
Message Model		
Status report size	256 B	Position, velocity, health
Coordination msg size	512 B	Task assignment, orbit adj.
Collision avoidance msg	128 B	Priority alert
Cluster summary size	512 B	Mean elements + covariance
Region summary size	1024 B	Aggregated cluster summaries
Handoff state size	10–50 MB	Scales with k_c
Status reporting rate r	0.1 msg/s	Per node
Collision avoidance rate	10^{-4}/node/s	Poisson; see text[a]
Link and Processing		
Per-node bandwidth	1 kbps	Coordination channel
Optical ISL capacity	1–10 Gbps	For handoff transfers
Processing capacity μ_s	1,000 msg/s	Per server (centralized)[b]
Cluster coord. μ_c	200 msg/s	Pipeline capacity[c]
Regional coord. μ_r	500 msg/s	Pipeline capacity[c]
Clusters per region k_r	$\lceil N/(k_c \cdot n_r) \rceil$	Derived; see text
Regional coordinators n_r	10	Default; scales with N
Message buffer size	10,000 msg	Drop-tail on overflow
Processing delay	Deterministic 5 ms	CPU time per message[c]
Gilbert-Elliott Link Model		
Good-state loss p_G	0.01	
Bad-state loss p_B	0.90	
Good→Bad prob. p_{GB}	0.05/cycle	Transition per T_c
Bad→Good prob. p_{BG}	0.50/cycle	Steady-state avail. 91%
Retransmissions M_r	2	Intra-cycle only
Failure Model		
Node failure rate	2%/year	Exponential distribution
MTTF per node	50 years	Operational phase only
Coordinator failure	Same as node	No enhanced reliability
Monte Carlo Configuration		
Runs per config	30	All configurations
Simulation duration	1 year	Per run
Clock resolution	1 s / 10 s	Collision / routine (T_c)
Random seed strategy	Sequential	Mersenne Twister

[a]Screening notifications (conjunction alerts requiring ground follow-up), not autonomous maneuver commands. [b]Set low for single-server bound; see the $M/D/c$ formula ($N_{\max} = c \cdot \mu_s / r$). [c]$\mu_c = 200$ msg/s (5 ms/msg including integrity check); $\mu_r = 500$ msg/s (2 ms); both single-threaded.

TABLE IV
PER-NODE BANDWIDTH BREAKDOWN AT $N = 100{,}000$ (ANALYTICAL ESTIMATE)

Component	Central. (ref.)	Hierarch.	G.-S. Mesh (ref.)
Status reports[a]	205 bps	205 bps	205 bps
Commands (512 B)[c]	∼410 bps	∼410 bps	—
Heartbeats (64 B)	—	∼51 bps	—
Aggregation[d]	—	∼4 bps	$O(N)$
Collision alerts	∼0.1 bps	∼0.1 bps	—
Total per node	∼615 bps	∼870 bps	>1 kbps
Protocol overhead[b]	∼41%[e]	∼46%	Exceeds

[a]Baseline (20.5%); excluded from η. [b]Beyond baseline: $\eta = \eta_0 + \eta_{\text{cmd}}$; architecture-specific $\eta_0 \approx 5\%$ (heartbeats + summaries); workload-dependent $\eta_{\text{cmd}} \approx 41\%$ (stress-case commands). [c]Stress-case: 512 B/node/cycle $\approx$ 406 bps, topology-invariant (broadcast Type 1: one TX, all listen; per-node unicast Type 2 requires 22-cycle stagger). [d]Summaries: <1% of η. [e]Centralized η is η_{cmd} only (no heartbeats/summaries); the 5 pp difference from hierarchical is η_0.

E. Communication Overhead Definition

Peak vs. average rate distinction. $C_{\text{node}} = 1$ kbps is a traffic budget; the coordinator's PHY operates at ≥24 kbps under TDMA. The 1 kbps RF backup applies during optical outages (<1% of lifetime) but is design-driving: it determines coordination survivability, PHY rate, and safe-mode policy. Results scale linearly with C_{node}; at ≥10 kbps all workloads are single-cycle feasible.

Canonical overhead definitions. All overhead quantities in this paper follow a strict three-tier decomposition:

- **Baseline telemetry** ($B_{\text{status}} = 256$ B/cycle $= 204.8$ bps, 20.5%): periodic ephemeris status reports. *Always present, topology-invariant, excluded from η.*
- **Architecture-specific overhead** ($\eta_0 \approx 5\%$): heartbeats (64 B/cycle, 5.1%), coordinator summaries (512 B/coordinator/cycle, 0.4% per node at $k_c = 100$), and election traffic ($< 0.1\%$, amortized). Present only in hierarchical topology. *Included in η.*
- **Workload-dependent overhead** (η_{cmd}): commands (512 B, ∼41% at stress), collision alerts (128 B, $< 0.1\%$). Gated by campaign duty factor d. *Included in η.*

The protocol overhead metric is therefore:

$$\eta = \eta_0 + d \cdot \eta_{\text{cmd}}, \quad \eta_{\text{total}} = \eta + 20.5\% \tag{3}$$

where $d \in [0, 1]$ is the campaign duty factor (Section IV-E). Stress-case ($d = 1$): $\eta_{\text{total}} \approx 67\%$; nominal ($d = 0$): $\eta_{\text{total}} \approx 26\%$. At $\gamma \in [0.7, 0.9]$, stress-case exceeds Slotted ALOHA capacity, confirming TDMA is required. **Loss/miss taxonomy:** (i) *link loss*: PHY-layer erasure (GE); (ii) *queue drop*: coordinator ingress overflow; (iii) *deadline miss*: after cycle cutoff (Table XI).

1) Coordinator Link Capacity Parameterization: We parameterize coordinator ingress capacity C_{coord} (kbps) rather than assuming full pooling. Egress uses a separate optical ISL. We evaluate $C_{\text{coord}} = \beta \cdot k_c$ kbps for $\beta \in \{0.01, 0.1, 0.25, 0.5, 1.0\}$; results in Section IV-A.

Table IV provides an approximate per-node bandwidth breakdown for each topology at $N = 100{,}000$. The fleet-level DES overhead ($\eta \approx 46\%$, Table XIV) matches this analytical accounting to within 0.05%.

F. Traffic Accounting

Protocol overhead η follows the canonical decomposition (Eq. 3). Optical-ISL state transfers (10–50 MB handoffs) are excluded from η as they do not traverse the RF-backup channel.

G. Performance Metric Definitions

- **Communication overhead** (η): $\hat{B}_{\text{DES}}/(N \times C_{\text{node}})$, measured from full-participation DES byte counts.
- **Coordination cycle** ($T_c = 10$ s): interval between state synchronization rounds, consistent across all topologies.
- **Coordination success**: *per-message delivery rate* (fraction received) and *per-cycle completion* (all k_c messages delivered within T_c; $\approx (1 - p_{\text{loss}})^{k_c}$).
- **System availability**: fraction of time each node has a functioning coordination path.

- **Message latency**: end-to-end delay including propagation, queueing, and processing.
- **Handoff**: Raft election [27] ($\sim$12.8 KB) plus state transfer (10–50 MB over optical ISL, 80–400 ms, excluded from η).

IV. Results

Instantiation parameters. All numeric results in Sections IV-A through IV-I are conditioned on the parameter set in Table III. The sizing equations (Section V-C) accept arbitrary parameters; practitioners substitute deployment-specific values.

Roadmap. The results are organized around two feasibility layers: message-layer byte budget (Layer 1) and half-duplex TDMA airtime schedulability (Layer 2). Sections IV-A through IV-C address key mechanisms—coordinator capacity sizing (IV-A), Age-of-Information coordination quality (IV-B), and correlated loss characterization (IV-C)—each with analytical cross-checks. Section IV-D verifies that these mechanisms interact independently under joint conditions. Section IV-E quantifies the workload design envelope. Sections IV-F through IV-I present verification, topology baselines, and parameter sensitivity.

A. Coordinator Capacity Sizing

The binding bottleneck is *cluster coordinator ingress*: $k_c - 1$ members (excluding the coordinator itself) each send 256 B/cycle, requiring $\approx$20.3 kbps for $k_c = 100$. Regional ingress and command downlink are not capacity-constrained (Table III). The coordinator's CPU ($\mu_c = 200$ msg/s) is not binding ($(k_c - 1) \times s_{\text{proc}} = 495$ ms $< T_c$); only the link-rate constraint (C_{coord}) causes drops.

Primary sizing (TDMA). Under deterministic TDMA slot assignment, each member occupies exactly one slot per cycle; the required PHY rate is $C_{\text{TDMA}} = (k_c - 1)S_{\text{eph}} \times 8/(T_c \times \gamma)$ (Eq. 5): $\approx$27 kbps at standards-derived $\gamma = 0.76$ ($k_c = 100$). **Slot-level verification.** A TDMA simulator ($k_c = 100$, 10,000 cycles) schedules every ingress and egress slot explicitly, including guard intervals, half-duplex turnaround, and GE-correlated losses. Under nominal (no loss): simulated ingress = 9,177 ms (matching the analytical budget exactly), margin = 614 ms, zero deadline misses. Under GE losses ($p_{BG} = 0.50$, $p_B = 0.90$) with $M_r = 1$ intra-cycle retransmission: mean ingress grows to $\sim$9,811 ms; 52.7% of cycles miss the T_c deadline—confirming intra-cycle ARQ is infeasible at 24 kbps (Section IV-C). At the recommended 30 kbps design point: margin = 2,396 ms, zero deadline misses even with $M_r = 1$.

Phase-staggered scheduling ($\phi_j = (j/n_{\text{clusters}}) \times T_c$) spreads the regional ingress burst, reducing drops (Fig. 2).

TDMA frame model. We derive γ from an explicit slot structure rather than assuming a fixed value. Each TDMA slot contains: preamble/sync (32 bits), header (16 bits), payload ($S_{\text{eph}} \times 8 = 2{,}048$ bits), CRC-16 (16 bits), and guard interval. At 24 kbps PHY rate, the data portion occupies $2{,}112/24{,}000 = 88.0$ ms. Guard time comprises propagation uncertainty and half-duplex turnaround: at 500 km cluster diameter, differential propagation is $\sim$1.7 ms; turnaround for CCSDS Proximity-1 class radios is $\sim$2 ms [28]; adding 1 ms timing jitter margin gives $T_{\text{guard}} = 4.7$ ms. Total slot duration: $88.0 + 4.7 = 92.7$ ms, yielding:

$$\gamma = \frac{T_{\text{data}}}{T_{\text{slot}}} = \frac{88.0}{92.7} = 0.949 \tag{4}$$

This slot-level value (0.949) does not include FEC coding ($\sim$12.5% at rate 7/8 LDPC), antenna acquisition ($\sim$5 ms dwell), or ranging/calibration. Section IV-I derives $\gamma = 0.76$ from CCSDS Proximity-1 framing, which we adopt as the standards-derived design value throughout this paper. The zero-drop raw capacity at $\gamma = 0.76$ is:

$$C_{\text{TDMA}} = \frac{(k_c - 1) \times S_{\text{eph}} \times 8}{T_c \times \gamma} \approx 26.7 \text{ kbps } (\gamma = 0.76) \tag{5}$$

Fleet-wide TDMA cost is 0.28 kbps/node (1% coordinators at $k_c = 100$); see Fig. 5. **Recommended design point:** 30 kbps provides $\sim$12% margin over $C_{\text{TDMA}} = 26.7$ kbps for ranging ($\sim$50 ms) and control-plane traffic not explicitly modeled. At 30 kbps: ingress to $\sim$9.1 s, margin $\sim$900 ms—tight but positive with the standards-derived γ.

MAC efficiency sensitivity. The γ parameter absorbs all physical-layer effects not explicitly modeled. Table VI shows how margin and schedulability degrade as γ decreases. At 24 kbps, schedulability requires $\gamma \geq 0.90$; at 30 kbps, $\gamma \geq 0.75$ suffices. Section IV-I derives $\gamma = 0.76$ from CCSDS Proximity-1 framing, placing the design point in the $\gamma = 0.75$–0.80 row—feasible at 30 kbps but infeasible at 24 kbps, consistent with Table VI.

1) Link Budget Justification: Table V derives achievable data rates for two operating modes from standard link budget parameters. The ISL mode (S-band, 2.2 GHz, 1 W, 6 dBi helical antennas, 500 km) provides >200 kbps maximum rate—far exceeding the 24–30 kbps coordinator requirement. The RF-backup mode (UHF, 400 MHz, 0.1 W, 0 dBi omnidirectional, 1000 km cluster-scale range) achieves a maximum rate of $\sim$2.5 kbps, justifying the 1 kbps design point as a conservative allocation for an always-available omnidirectional backup radio. Higher rates would require directional antennas, contradicting the "always-available backup" operational concept.

Scaling to $\geq$10 kbps. At $C_{\text{node}} = 10$ kbps: coordinator PHY = 240 kbps, ingress = 920 ms, margin = 9,060 ms ($> 90\%$ of T_c)—both byte-budget and schedulability trivially satisfied.

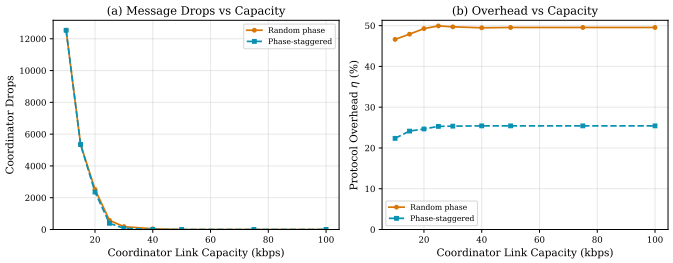

Fig. 2. Phase-stagger experiment ($N = 10^4$, $k_c = 100$). (a) Coordinator drops vs. link capacity: phase staggering eliminates drops at $\sim$25 kbps vs. 50 kbps under random phase. (b) Overhead is unaffected by scheduling mode when capacity is sufficient.

TABLE V
LINK BUDGET: ISL VS. RF-BACKUP OPERATING MODES

Parameter	ISL Mode	RF-Backup
Frequency	2.2 GHz (S-band)	400 MHz (UHF)
Tx power	1.0 W (0 dBW)	0.1 W (−10 dBW)
Tx/Rx gain	6 dBi (helical)	0 dBi (omni)
Distance	500 km	1000 km
FSPL	153.3 dB	144.5 dB
P_{rx}	−141.3 dBW	−154.5 dBW
E_b/N_0 @ 1 kbps	29.7 dB	16.5 dB
E_b/N_0 @ 24 kbps	15.9 dB	2.7 dB (fails)
Max rate (BPSK, BER$< 10^{-5}$)	>200 kbps	∼2.5 kbps

Assumptions: $T_{\text{sys}} = 290$ K, NF = 3 dB, implementation loss = 2 dB, required $E_b/N_0 = 9.6$ dB (BPSK, BER $< 10^{-5}$).

TABLE VI
MAC EFFICIENCY (γ) SENSITIVITY: MARGIN AND SCHEDULABILITY ($k_c = 100$, NO LOSS, SLOT-LEVEL SIMULATION, 10,000 CYCLES)

	Guard	24 kbps		30 kbps	
γ	(ms)	Margin (ms)	Sched.	Margin (ms)	Sched.
0.60	58.7	−4,837	N	−1,870	N
0.65	47.4	−3,697	N	−958	N
0.70	37.7	−2,721	N	−177	N
0.75	29.3	−1,874	N	500	Y
0.76[b]	**27.8**	**−1,681**	**N**	**693**	**Y**
0.80	22.0	−1,133	N	1,093	Y
0.85	15.5	−480	N	1,616	Y
0.90	9.8	101	Y	2,081	Y
0.949[a]	4.7	611	Y	2,488	Y

[a]Derived from slot structure (Eq. 4). Guard time computed as $T_{\text{data}} \times (1/\gamma - 1)$. The standards-derived design value is $\gamma = 0.76$ (Section IV-I), which is feasible at 30 kbps but infeasible at 24 kbps. [b]Standards-derived from CCSDS Proximity-1 (Table XVIII).

2) Fleet-Level Channel Reuse: With F orthogonal RF channels and spatial reuse factor R, the maximum simultaneously active clusters is $F \times R$. If f_{RF} is the fraction of the fleet in RF-backup mode, the number of active RF-backup clusters is $N_{\text{RF}} = \lceil f_{\text{RF}} \cdot N/k_c \rceil$, requiring time-share groups:

$$G = \left\lceil \frac{N_{\text{RF}}}{F \times R} \right\rceil, \quad T_c^{\text{fleet}} = \max(T_c,\ G \cdot T_c) \tag{6}$$

The per-cluster equations are exact (no AoI inflation) when $f_{\text{RF}} \leq F{\cdot}R{\cdot}k_c/N$. For $F = 4$, $R = 3$, $k_c = 100$, $N = 10^5$: the non-binding threshold is $f_{\text{RF}} \leq 1.2\%$. Since ISL outages are expected to affect <1% of the fleet under normal operations, per-cluster sizing holds without modification.
When $G > 1$, AoI P99 scales as $G \times 440$ s. At $f_{\text{RF}} = 5\%$: $G = 5$, $T_c^{\text{fleet}} = 50$ s, AoI P99 = 2,200 s—acceptable for safe-mode (Fig. 3).

Half-duplex TX/RX partitioning. The coordinator partitions T_c between ingress (99×92.7 ms = 9.18 s, 91.8%) and egress (∼200 ms: one 512 B broadcast command at 171 ms + 64 B heartbeat at 21 ms). Two command types: *Type 1 (broadcast):* 171 ms, single cycle. *Type 2 (per-node unicast):* $k_c \times 171$ ms = 16.9 s $> T_c$, requires multi-cycle staggering.

Command generation locus. The above assumes centralized command generation, making η_{cmd} topology-invariant. Cluster-local Raft consensus [27] yields $\eta_{\text{consensus}} = 3.1\%$ per decision (3,840 B: proposal + quorum + commit), replacing

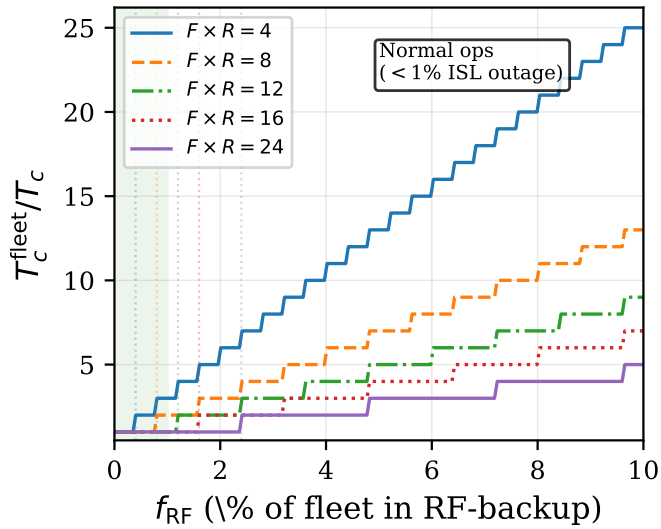

Fig. 3. Fleet-level coordination period inflation vs. RF-backup fraction f_{RF} for various frequency × spatial reuse products $F \times R$. Normal operations ($f_{\text{RF}} < 1\%$, green shading) are non-binding for all configurations. Correlated outages ($f_{\text{RF}} = 5$–10%) inflate T_c proportionally.

the 41% centralized command overhead with substantially lower topology-dependent overhead. The sizing equations adapt by substituting the appropriate S_{cmd} and p_{cmd} for each architecture.

Schedulability vs. byte budget. The overhead metric η counts information content per node per cycle, not PHY delivery time. Under Type 1 (broadcast), egress fits in one cycle (171 ms; Eqs. 9–10). Under Type 2 (per-node unicast), egress requires $k_c \times 171$ ms = 16.9 s $> T_c$:

$$L_{\text{cmd}} = \left\lceil \frac{k_c \cdot T_{\text{cmd}}}{T_c \cdot (1 - \alpha_{\text{RX}})} \right\rceil = \left\lceil \frac{16.9}{0.8} \right\rceil = 22 \text{ cycles} \tag{7}$$

More generally, if fraction q of stress-case commands require per-node unicast addressing, the stagger requirement generalizes to:

$$L_{\text{cmd}}(q) = \left\lceil \frac{(1 + \lfloor q \cdot k_c \rfloor) \cdot T_{\text{cmd}}}{T_c \cdot (1 - \alpha_{\text{RX}})} \right\rceil \tag{8}$$

Figure 4 shows $L_{\text{cmd}}(q)$ at 24 kbps and 30 kbps; slot-level simulation confirms the analytical curve. At $q \leq 0.05$ (5% unicast), $L \leq 2$ cycles; at $q = 0.5$, $L = 11$ cycles. The byte budget η is independent of q; only Layer 2 airtime schedulability changes.
Table XII maps each workload profile to its schedulability status. Safety-critical commands (128 B broadcast, 43 ms egress) are always single-cycle feasible; the 22-cycle stagger applies only to per-node unicast commands.

TDMA frame-time feasibility. The TDMA frame imposes time-slot constraints alongside byte-rate constraints:

$$\text{Ingress:} \quad (k_c - 1) \cdot T_{\text{slot}} \cdot (1 + \bar{M}_r) \leq T_c \cdot \alpha_{\text{RX}} \tag{9}$$

$$\text{Egress:} \quad T_{\text{cmd}} + T_{\text{hb}} + T_{\text{sync}} \leq T_c \cdot (1 - \alpha_{\text{RX}}) \tag{10}$$

where α_{RX} is the ingress fraction of T_c and $\bar{M}_r$ is the retransmission fraction. Under nominal conditions ($\bar{M}_r = 0$, broadcast): ingress 9,177 ms + egress 200 ms = 9,377 ms $< T_c$. Under GE steady-state, the expected fraction of slots

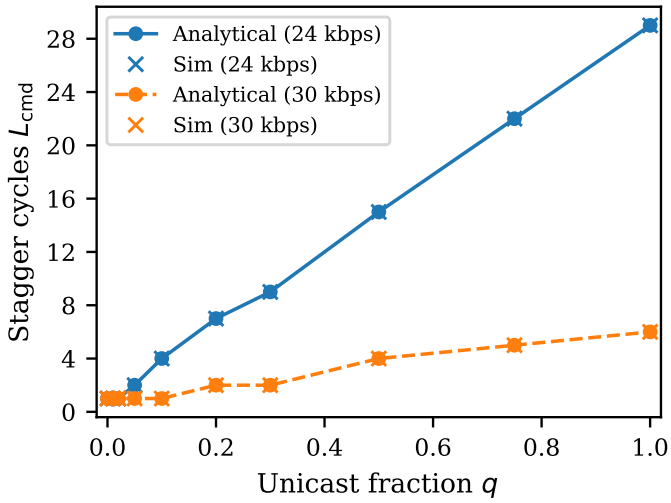

Fig. 4. Command stagger cycles L_{cmd} vs. unicast fraction q (Eq. 8). Analytical curves at 24 kbps and 30 kbps with slot-level simulation validation (crosses). At $q = 0$: broadcast, single cycle. At $q = 1$: full unicast, 22 cycles (24 kbps) or 15 cycles (30 kbps). Mixed addressing ($q \leq 0.05$) keeps stagger ≤ 2 cycles.

TABLE VII
TDMA SUPERFRAME TIME BUDGET ($k_c = 100$, 24 KBPS PHY)

Component	Duration (ms)	Direction
Ingress (RX)		
Member report slots (99 × 92.7 ms)	9,177	RX
Sync beacon (8 bits at 24 kbps)	0.3	TX
Egress (TX)		
Command broadcast (512 B)	171	TX
Heartbeat (64 B)	21	TX
TX/RX turnaround (×2)	4	—
Re-sync preamble	4	TX
Total allocated	**9,377**	
Unallocated margin	**623**	

Margin covers ranging (∼50 ms), clock drift (<0.6 ms), and contingency. Unicast stagger (Eq. 7) uses net egress = 0.80 s. Degraded $\gamma = 0.80$: ingress = 9,702 ms, margin = 98 ms. Slot-level sim confirms these values exactly (Section IV-A).

requiring retransmission is $\pi_B \cdot p_B/(\pi_B + \pi_G) \approx 0.08$; the resulting retransmission airtime is $(k_c - 1) \times 0.08 \times T_{\text{slot}} \approx$ 740 ms, which alone exceeds the 623 ms margin (Table VII). Intra-cycle ARQ is therefore infeasible; recovery relies on inter-cycle repetition (Section IV-C).

TDMA synchronization. GPS/GNSS (<100 ns) with TCXO holdover (<0.6 ms/10 min). Under GNSS denial: sync beacon (0.8% of η); if lost, Slotted ALOHA ($\gamma \approx 0.36$, $\eta_{\text{total}} \approx 5.7\%$).

Complete superframe time budget. Table VII itemizes all time allocations within the $T_c = 10$ s cycle, quantifying the margin available for unmodeled overheads.

$$C_{\text{raw}} = C_{\text{coord}}/\gamma \tag{11}$$

B. Coordination Quality: Age of Information

We adopt the Age of Information (AoI) framework [41]–[43]: $\text{AoI}_{ij}(t) = t - t_{\text{last}}(i, j)$. Under periodic reporting, mean

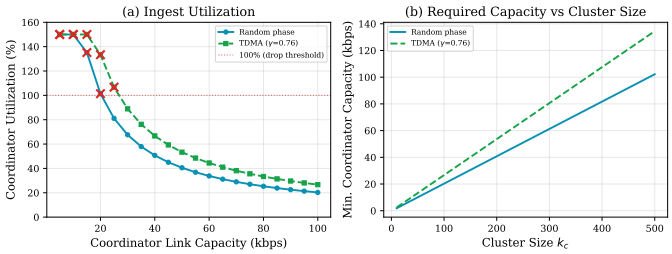

Fig. 5. TDMA scheduling analysis. (a) Coordinator utilization vs. link capacity. (b) Minimum coordinator link capacity vs. cluster size.

TABLE VIII
AGE OF INFORMATION AT CLUSTER COORDINATORS ($N = 10^4$, $k_c = 100$)

Configuration	Mean AoI (s)	P99 AoI (s)	Max AoI (s)	η (%)
Exception telemetry ($p_{link} = 1.0$, no link loss)				
Full reporting ($p_{\text{exc}} = 1.0$)[a]	4.9	10.0	10.0	46.0
$p_{\text{exc}} = 0.10$	47	441	780	4.9
$p_{\text{exc}} = 0.30$	28	129	355	14.0
$p_{\text{exc}} = 0.50$	15	59	147	23.2
Link availability (full reporting, $p_{exc} = 1.0$)				
$p_{\text{link}} = 1.0$	4.9	10.0	10.0	46.0
$p_{\text{link}} = 0.8$	7.4	30	67	36.8
$p_{\text{link}} = 0.5$	13	66	175	23.0

DES-measured AoI sampled every 100 s across all coordinators. [a]Full reporting: all nodes transmit every cycle; $\eta = 46\%$ includes stress-case commands + heartbeats + summaries (this is the stress workload from Table XII, not baseline telemetry alone). **Tail-statistic methodology:** per-run P99 computed over $\sim 3.15 \times 10^6$ samples; reported P99 is the mean of 30 per-run values with 95% bootstrap CI.

AoI $\approx T_c/2 = 5$ s. Exception telemetry degrades AoI proportionally: at $p_{\text{exc}} = 0.10$, P99 AoI exceeds 440 s, motivating future coupling to orbital prediction models (Section V-A).

Analytic cross-check. Under exception reporting, inter-report intervals follow $\text{Geom}(p_{\text{exc}})$, giving:

$$\text{AoI}_{\text{P99}} = T_c \cdot \left\lceil \frac{\ln(0.01)}{\ln(1 - p_{\text{exc}})} \right\rceil \tag{12}$$

For $p_{\text{exc}} = 0.10$: $\text{AoI}_{\text{P99}} = 440$ s, consistent within one T_c step with the DES value of 441 s (bootstrap 95% CI: $[438, 444]$ s, $n = 30$).

TDMA and mission coupling. AoI measures inter-update gap (not single-message latency); TDMA slots do not affect the inter-cycle metric. Table IX maps AoI to operational functions. ESA TCAs are 12–72 h [2]; 441 s is $< 0.5\%$ of a 24 h window. Final maneuvers require AoI < 10 s via optical ISL.

C. Correlated Loss Characterization

Under i.i.d. losses with $M_r = 2$ retransmissions, $p_{\text{success}} = 1 - (1 - p_{\text{link}})^3$ (87.5% at $p_{\text{link}} = 0.5$). To test correlated losses, we implement a Gilbert-Elliott (GE) model: good state ($p_G = 0.01$), bad state ($p_B = 0.90$), with transition probabilities $p_{GB} = 0.05$ and $p_{BG} = 0.50$. **Coherence assumption:** the GE state transitions once per $T_c = 10$ s cycle; intra-cycle retransmissions all face the same channel state. This is a *modeling choice*, not a physical property—it

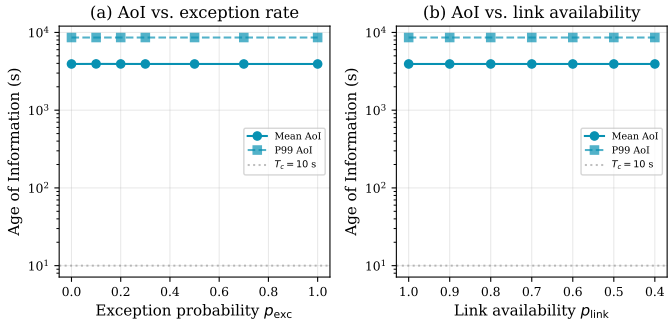

Fig. 6. AoI at coordinators. (a) Geometric growth with decreasing p_{exc}. (b) Link losses modestly degrade AoI.

TABLE IX
AoI Operational Mapping

Function	Required AoI	Channel	Satisfied?
Conjunction screening	< 24 h	RF backup	Yes (440 s)
Maneuver execution	< 10 s	Optical ISL	Yes ($T_c/2 = 5$ s)
Formation keeping	< 60 s	Optical ISL	Yes
Safe-mode exception	$\leq$ P99	RF backup	By design (440 s)

determines whether intra-cycle ARQ faces correlated or independent fading. The per-cycle model ($\tau_c = T_c$) is conservative for recovery: structural shadowing ($\tau_c \approx$ 1–10 s) matches T_c, making the assumption approximately correct; antenna mispointing ($\tau_c \approx$ 10–60 s) has $\tau_c \geq T_c$, making the per-cycle model optimistic for burst *duration* (real bursts may span multiple cycles, requiring the $p_{BG} \leq 0.10$ design curves in Fig. 8b). Shorter coherence ($\tau_c < T_c$) would allow mid-cycle state transitions, enabling conventional ARQ (Fig. 7). Under bad-state bursts, all $M_r + 1 = 3$ attempts face $p_B = 0.90$, yielding only 27.1% recovery (vs. 87.5% i.i.d.)—confirming intra-cycle retransmission is structurally ineffective during correlated bursts.

Inter-cycle recovery (DES-validated). Per-member loss streaks tracked across cycles. Figure 8(a): DES vs. Markov-chain CDF at $p_{BG} = 0.50$. Mean recovery 1.7 cycles, P95 = 4 cycles; >98% recovered by 5 cycles. Buffer requirement: ~77 kB per coordinator.

Markov recovery derivation. Starting from bad state: $\pi_G^{(k)} = p_{BG}\pi_B^{(k-1)} + (1 - p_{GB})\pi_G^{(k-1)}$, $\pi_B^{(0)} = 1$. Recovery at cycle k: $p_{\text{rec}}^{(k)} = \pi_G^{(k)}(1 - p_G^{M_r+1}) + \pi_B^{(k)}(1 - p_B^{M_r+1})$. CDF: $F(k) = 1 - \prod_{j=1}^{k}(1 - p_{\text{rec}}^{(j)})$; P95 $= \min\{k : F(k) \geq 0.95\}$.

GE parameter sensitivity. Figure 8(b) shows how P95 recovery varies with p_{BG} for three bad-state loss rates ($p_B \in \{0.70, 0.90, 0.95\}$). Recovery times are dominated by p_{BG}: at $p_{BG} = 0.50$ (mean 2-cycle outage bursts), P95 = 3–5 cycles across all p_B; at $p_{BG} = 0.10$ (mean 10-cycle bursts), P95 reaches 12–18 cycles. DES validation points (squares) confirm the analytical curves. This provides a family of design curves allowing practitioners to size buffers and AoI tolerances for arbitrary channel burstiness.

Physical mapping and model scope. The GE analysis is a *parametric sensitivity study*: it characterizes recovery behavior as a function of (p_{BG}, p_B) and provides design curves (Fig. 8b) from which practitioners select parameters matching their channel. The Lutz et al. two-state framework [52] was developed for land-mobile satellite channels (L/S-band terrestrial shadowing); ISL self-blockage from spacecraft appendages differs in mechanism but shares the two-state structure (clear/obstructed). No ISL-specific GE measurement data are available in the open literature; the default parameterization ($p_{GB} = 0.05$, $p_{BG} = 0.50$, within the Lutz range) should be treated as illustrative rather than predictive. ITU-R P.681 [53] provides related Markov statistics for Earth-space mobile links.

TABLE X
GE Mechanism→Parameter Mapping[a]

Mechanism	τ_c	p_{BG}	p_B	P95 Rec.
Structural shadowing	1–10 s	$\geq$0.50	0.70–0.90	3–5 cyc
Antenna mispointing	10–60 s	~0.10	0.90–0.95	12–18 cyc
Earth occultation	~35 min	Use visibility schedules		

[a]No ISL-specific GE measurement data are available; parameters are estimated from physical mechanisms. The sensitivity sweep (Fig. 8b) allows practitioners to read P95 for their measured channel.

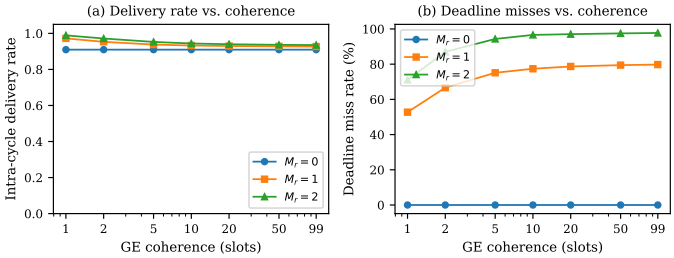

Fig. 7. Intra-cycle ARQ effectiveness vs. GE coherence time (slot-level simulation, $k_c = 100$, 10,000 cycles). (a) Delivery rate: fast-mixing channels (≤5 slots) recover >97% with $M_r = 2$; per-cycle coherence (99 slots) limits recovery to 93.5%. (b) Deadline miss rate: $M_r = 2$ at per-cycle coherence causes 97.7% misses (all retransmissions fall in same GE state); fast mixing reduces misses to 71%.

Three LEO obstruction mechanisms map to distinct GE regimes (Table X). **Decision rule:** if $\tau_c \geq T_c$, use inter-cycle recovery; if $\tau_c \approx$ slot duration, intra-cycle ARQ may be effective (Fig. 7).

Coherence-time sensitivity. Figure 7 varies coherence from per-slot (~93 ms) to per-cycle (~10 s). At per-slot coherence, delivery reaches 98.9% (vs. 93.5% per-cycle). The ARQ-infeasibility conclusion is specific to slow-mixing obstructions; fast-fading channels recover within the intra-cycle budget.

D. Joint Parameter Interaction Verification

We test whether loss model and retransmission policy interact with TDMA scheduling by sweeping PHY rate across four channel conditions in the slot-level simulator ($k_c = 100$, 10,000 cycles). Table XI reports deadline miss rates.

TDMA pipeline decoupling. Under $M_r = 0$, GE losses and TDMA scheduling are decoupled: columns 1–2 of Table XI are identical at every PHY rate (fixed slots regardless of delivery outcome). Under ARQ ($M_r \geq 1$), retransmission slots break decoupling at ≤24 kbps (52.7% misses)—a TDMA-specific interaction invisible to the fluid-server DES. At 30 kbps, even $M_r = 1$ produces zero misses.

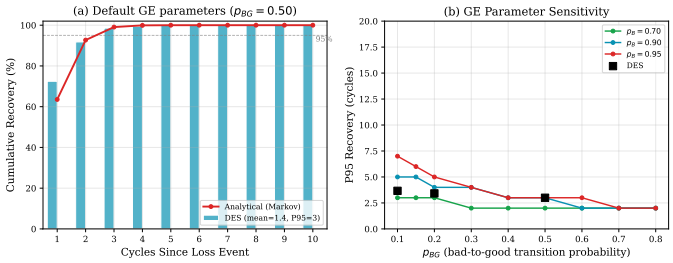

Fig. 8. Inter-cycle recovery under GE correlated loss. (a) CDF at default parameters ($p_{BG} = 0.50$, $p_B = 0.90$): DES bars (30 MC replications, $N = 10{,}000$, $k_c = 100$) vs. Markov-chain analytical model. (b) P95 recovery cycles vs. p_{BG} for three p_B values; DES validation at $p_{BG} \in \{0.10, 0.20, 0.50\}$ (squares, 30 MC replications each).

TABLE XI
TDMA-AWARE JOINT INTERACTION: DEADLINE MISS RATE (%) BY PHY RATE × CHANNEL CONDITION ($k_c = 100$, 10,000 CYCLES)

PHY Rate	No Loss	GE $M_r = 0$	GE $M_r = 1$	GE+Exc
15 kbps	100.0	100.0	100.0	100.0
20 kbps	100.0	100.0	100.0	100.0
24 kbps	0.0	0.0	52.7	0.0
25 kbps	0.0	0.0	7.6	0.0
30 kbps	0.0	0.0	0.0	0.0
50 kbps	0.0	0.0	0.0	0.0

Slot-level TDMA simulator with GE correlated losses ($p_G = 0.01$, $p_B = 0.90$, $p_{GB} = 0.05$, $p_{BG} = 0.50$). No Loss: $p_G = p_B = 0$. GE+Exc: GE $M_r = 0$ with exception telemetry ($p_{\text{cmd}} = 0$, reduced offered load). Columns 1–2 are identical: under $M_r = 0$ each member transmits in its fixed slot regardless of channel state, so loss and scheduling are decoupled.

E. Workload Design Envelope

Three workload profiles bound the operational design envelope (Table XII). **Per-node per-cycle messages:** all profiles include 1 status report (256 B) and 1 heartbeat (64 B). **Nominal (N):** $p_{\text{cmd}} = 0$ (no commands); $p_{\text{exc}} = 1.0$ (periodic). **Event (E):** $p_{\text{cmd}} = 0.01$; $p_{\text{exc}} = 0.10$. **Stress (S):** $p_{\text{cmd}} = 1.0$ (512 B command every cycle, broadcast Type 1 or unicast Type 2); $p_{\text{exc}} = 1.0$. The event-driven profile ($\eta_E \approx 6\%$) is representative of routine operations; the stress-case ($\eta_S \approx 46\%$) bounds continuous fleet-wide reconfiguration. In practice, such campaigns are episodic; the campaign duty factor d (Table XIII) parameterizes this. Fig. 9 confirms all three profiles produce scale-invariant η across $N = 10^3$ to 10^5.

Campaign duty factor. The stress-case assumes continuous commands ($d = 1$, every cycle). In practice, maneuver campaigns (orbit-raising, collision avoidance) are episodic. A campaign duty factor $d \in [0, 1]$ gates command generation per cycle (Bernoulli with probability d), yielding $\eta(d) = \eta_0 + d \cdot \eta_{\text{cmd}}$ (Eq. 3). Table XIII shows η across the operational range. For routine operations ($d = 0.01$), overhead is negligibly above nominal; even intensive reconfiguration ($d = 0.10$) adds only $\sim$4%. The stress-case ($d = 1$) remains useful as a strict upper bound.

Representative campaign scenarios. Three worked examples anchor d in concrete mission phases: (1) *Orbit-raising* ($d = 0.05$, $L_{\text{on}} = 500$ cycles/83 min, 2×7 d/yr): annual average $\eta \approx 7\%$, peak $\eta = 7.1\%$. (2) *Station-keeping* ($d = 0.01$, $L_{\text{on}} = 50$ cycles/8 min, weekly): $\eta \approx 5.4\%$, peak $\eta = 5.4\%$. (3) *Collision avoidance* ($d \to 1.0$ during event, $L_{\text{on}} = 6$ cycles/1 min, $\sim$12 events/yr): background $\eta \approx 5\%$, peak $\eta = 46\%$ (stress bound). Routine operations thus occupy $\eta \approx 5$–10%; the 46% stress-case is episodic (<1% of operational time).

TABLE XII
WORKLOAD FEASIBILITY ($k_c = 100$, 24–30 KBPS HALF-DUPLEX, $\gamma = 0.76$)

	Layer 1: Byte Budget			Layer 2: Airtime Schedulability	
Profile	η	η_{total}[b]	$\eta_{\text{total}}/\gamma$[c]	**1-Cycle?**	**Cycles**
Nominal (N)	5%	26%	34%	✓	1
Event (E)	6%	27%	36%	✓	1
Stress (bcast)	46%	67%	88%	✓	1
Stress (unicast)	46%	67%	88%	—	22[a]

Reading: Layer 1 checks the message-layer byte budget (η, η_{total}). The PHY translation $\eta_{\text{total}}/\gamma$ is a necessary condition for schedulability (not an independent layer); Layer 2 (half-duplex TDMA airtime) is the decisive test: can ingress + egress fit in a single superframe within T_c? Stress unicast passes byte budget but fails half-duplex airtime. [a]$L_{\text{cmd}} = \lceil k_c T_{\text{cmd}}/(T_c - T_{\text{ingress}}) \rceil = 22$ cycles (Eq. 7). [b]$\eta_{\text{total}} = \eta + 20.5\%$ baseline. [c]PHY translation: necessary but not sufficient for Layer 2.

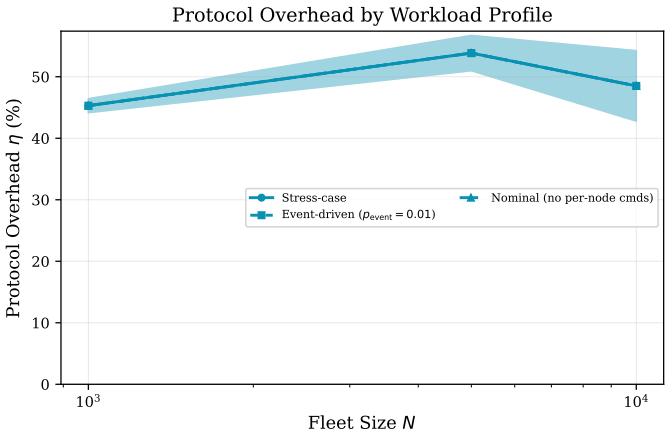

Fig. 9. Hierarchical overhead across three workload profiles. The 9× stress-to-nominal spread bounds the design envelope.

Temporal correlation. The Bernoulli model assumes i.i.d. duty-factor draws; real campaigns exhibit temporal correlation. We implement an ON/OFF Markov alternative (geometric ON duration L_{on} cycles, OFF duration $L_{\text{off}} = L_{\text{on}}(1 - d)/d$) that preserves marginal d but produces temporally correlated bursts. Fig. 10 confirms that ON/OFF ($L_{\text{on}} = 100$) produces heavier tails than Bernoulli at identical $d = 0.10$: the P99 coordinator ingress increases by $\sim$15%, relevant for buffer sizing. The Bernoulli model is exact for mean η and optimistic for peak buffer occupancy.

η_0 audit. η_0 components: heartbeats 5.1% [31], summaries 0.4%, election < 0.1% (sum: 5.6%); DES measures 5.0%. The 0.5 pp gap: coordinators (1% of fleet at $k_c = 100$) do not send status/heartbeat to themselves; fleet-average saving $= 0.01 \times (256 + 64) \times 8/(1000 \times 10) = 0.026$ pp per component, ≈ 0.5 pp total accounting for summary self-exclusion. Commands account for >60% of stress-case traffic, summaries <1%.

TABLE XIII
OVERHEAD VS. CAMPAIGN DUTY FACTOR ($k_c = 100$, $\gamma = 0.76$ AT 24 KBPS REFERENCE)

d	η	η_{total}	$\eta_{\text{total}}/\gamma$	TDMA?	Interpretation
0.00	5.0%	25.5%	33.6%	No	Nominal
0.01	5.4%	25.9%	34.1%	No	Routine ops
0.05	7.1%	27.6%	36.3%	No	Orbit-raising
0.10	9.1%	29.6%	38.9%	No	Intensive reconfig
0.50	25.5%	46.0%	60.5%	Yes	Extended campaign
1.00	46.0%	66.5%	87.5%	Yes	Continuous (bound)

$\eta(d) = \eta_0 + d \cdot \eta_{\text{cmd}}$; $\eta_{\text{cmd}} = 41\%$; baseline 20.5%. TDMA required when $\eta_{\text{total}}/\gamma > 50\%$ (half-duplex margin).

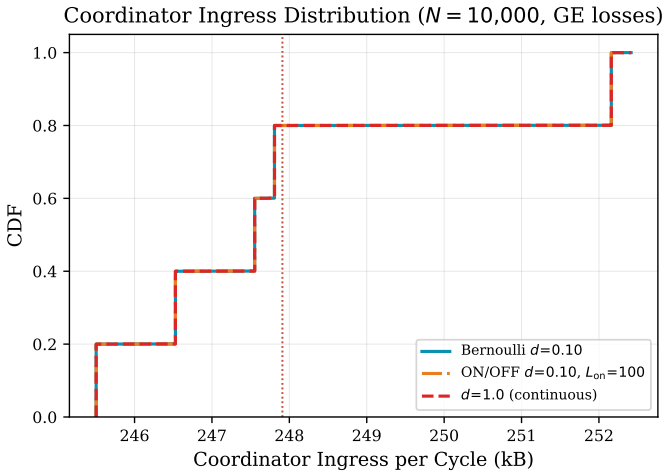

Fig. 10. CDF of per-cycle coordinator ingress bytes ($N = 10{,}000$, $k_c = 100$, GE losses). Bernoulli $d = 0.10$ (solid): bimodal, ~90% of cycles at baseline. ON/OFF $d = 0.10$, $L_{\text{on}} = 100$ (dash-dot): heavier tail from sustained bursts. Continuous $d = 1.0$ (dashed): all cycles at full load. Vertical lines: closed-form means.

Distributed planning. 3-round Raft consensus [27] adds $\eta_{\text{consensus}} \approx 30.7\%$, competing with centralized $\eta_{\text{cmd}} = 41\%$; the "topology-invariant" conclusion holds only under centralized broadcast semantics.

F. Implementation Consistency and Distributional Analysis

1) Scaling Behavior: DES matches closed-form to <0.1% at all fleet sizes 10^3–10^5 (Table XIV; implementation consistency, expected by construction). The DES's independent contribution is *distributional*: Fig. 10 shows that under stochastic duty-factor campaigns ($d = 0.10$) with GE burst losses, coordinator ingress variability is bimodal—most cycles near baseline, active cycles at full load—a property not captured by the closed-form mean.

2) Analytical Cross-Check: Exception telemetry reduces overhead linearly: $\eta(p_{\text{exc}}) \approx p_{\text{exc}} \times \eta(1.0)$ (verified at 6 configurations; all errors $\leq 1\%$; Fig. 11).

3) Parametric Sensitivity: Fig. 12 presents parametric sensitivity: overhead scales linearly with r; at $\gamma = 0.7$, hierarchical reaches ~66%; at $\gamma < 0.5$ (CSMA), stress-case exceeds capacity.

G. Topology Comparison and Baselines

Table XV and Fig. 13 summarize scaling behavior.

TABLE XIV
HIERARCHICAL COMMUNICATION OVERHEAD SCALING ($k_c = 100$)[A]

Fleet Size	η_{DES} (%)	η_{analytic} (%)	Δ (%)	η_{eff} (%)[b]
10^3–10^5[c]	46.0	46.1	<0.1	51–66
With exception-based telemetry ($p_{exc} = 0.30$)				
10^4–10^5	14.0	13.8	<1.5	16–20

[a]30 MC replications, SD < 0.001%. [b]$\eta_{\text{eff}} = \eta_{\text{DES}}/\gamma$ for $\gamma \in [0.7, 0.9]$. [c]$\eta = 46.0\%$ at all 10 fleet sizes (10^3 through 10^5); scale-invariant by construction.

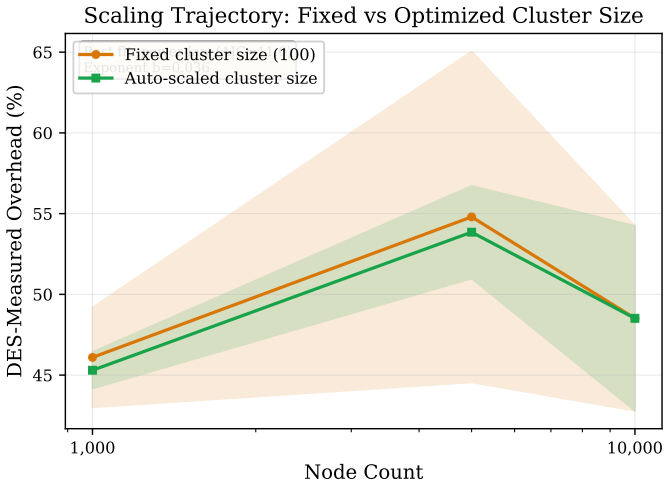

Fig. 11. Overhead trajectory (log-linear, DES-measured). Hierarchical $\eta \approx 46\%$ constant across 10^3–10^5. Dashed: exception telemetry.

Centralized baseline ($c = N/k_c$). Centralized processing does not diverge until $N \approx 10^6$ under $M/D/c$. This model captures compute-queue scalability only—not uplink scheduling or ground contact availability [9].

Latency. Within-cycle queueing: ~260 ms mean ($D[k_c]/D/1$ batch: propagation 1.7 ms + processing 5 ms/msg + batch wait $k_c s_{\text{proc}}/2$), P95 ≈ 500 ms. AoI: 44–441 s (Table VIII). Centralized round-trip is lower; the hierarchical advantage is fault tolerance.

Bandwidth–robustness tradeoff. The hierarchy saves bandwidth via $1/k_c$ aggregation but depends on coordinator availability (99.5%); the mesh provides local robustness at higher overhead with narrower scope (Table XVI). The hierarchy achieves $\gamma = 0.76$ via coordinator-assigned TDMA; decentralized architectures would require distributed TDMA coordination or CSMA/CA ($\gamma \approx 0.3$–0.5).

H. Cluster, Duty Cycle, and Link Sensitivity

1) Cluster Size: Protocol overhead is nearly invariant to k_c: $\eta = 46.0\%$ at $k_c \in \{50, 100, 200, 500\}$ ($\pm 0.1\%$, 30 MC replications). The trade-off is *latency*: $k_c = 50$ increases regional queueing to 675 ms (vs. 508 ms at $k_c = 100$). $k_c = 100$–200 is favorable (Fig. 14).

2) Coordinator Duty Cycle: The 24–48 h duty cycles occupy the Pareto frontier (Fig. 15): power CV 12–18%, handoff success $\geq 99.5\%$, system availability $\geq 99.2\%$ (two-state Markov, MTTF = 50 yr [24]). Shorter cycles (1–6 h)

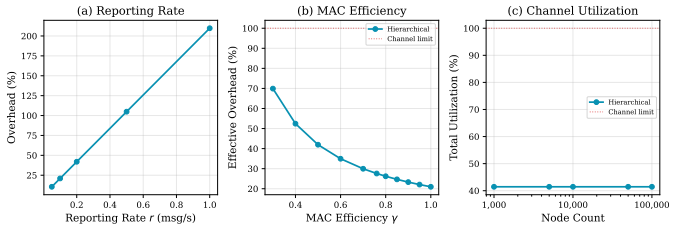

Fig. 12. Parametric sensitivity. (a) Overhead vs. reporting rate. (b) MAC efficiency γ vs. effective overhead. (c) Total channel utilization.

TABLE XV
COORDINATION TOPOLOGY COMPARISON

Architecture	Scalability Limit	O_{protocol}[a]	Failure M
Reference Bounds			
Centralized (c=1, bound)	∼10,000[b]	—[d]	Single point
Centralized (c=N/k_c)	∼10^6[c]	—[d]	Single point
Global-State Mesh (UB)	∼1,000	>100%	Distributed (99.99%)
Architecture Under Study			
Hierarchical	10^5 (validated)	5–46%	Graceful (99.5%)

[a]Protocol overhead beyond the topology-invariant 20.5% baseline. Hierarchical range: ∼5% (exception telemetry, $p_{\text{exc}} = 0.10$) to ∼46% (stress-case full reporting).
[b]Processing limit at $c = 1$; propagation/spectrum limits persist.
[c]Realistic provisioning ($c = N/k_c$, $\rho \leq 0.7$); binding constraint is uplink spectrum and latency, not processing. [d]Not applicable (compute-queue model only). A sectorized mesh (η = 65–67%, local-neighbor monitoring only, Section III-B4) is plotted in Fig. 13 for reference but is not functionally comparable.

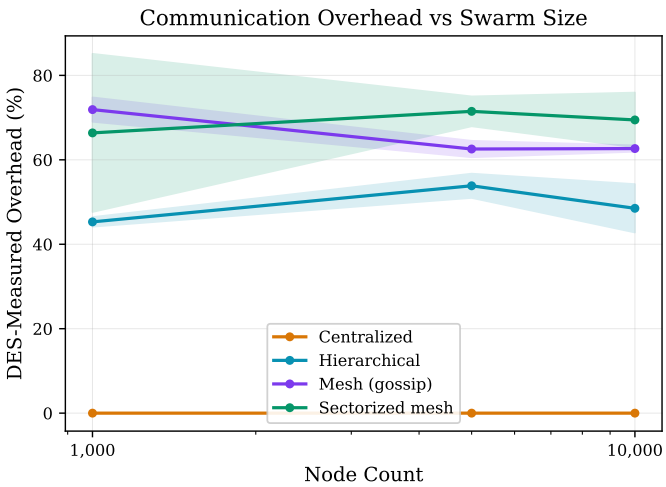

Fig. 13. Overhead vs. node count (log–log). Centralized (c=1) is *compute-queue only*—divergence reflects processing, not communications. Mesh saturates; hierarchical stays at ∼46%.

TABLE XVI
FUNCTIONAL CAPABILITY MATRIX

Function	Cent.	Hier.	Mesh	Global
Fleet command dissem.	✓	✓	—	—
Full cluster awareness	✓[a]	✓	—	✓
Local neighbor monitor	—	✓	✓	✓
Summary aggregation	✓[a]	✓	—	—
No coordinator depend.	—	—	✓	✓
Fits 1 kbps budget	—[b]	✓	✓	—

[a]At ground; not in-orbit autonomous. [b]Uplink-limited, not budget-limited.

increase handoff cost without improving availability; longer (7 d) degrades power uniformity to 35% CV. The 99.5% in Table XV conservatively accounts for cascading effects.

3) Link Availability: Table XVII summarizes coordination success under Bernoulli losses. Retransmission ($M_r = 2$) results are byte-budget delivery rates; TDMA airtime is not enforced. **Regime A** (≥10 kbps or $k_c \leq 30$): retransmissions fit in the superframe. **Regime B** (RF-backup, 24–30 kbps, $k_c = 100$): effective $M_r = 0$ intra-cycle; recovery is inter-cycle (Section IV-C).

I. Standards-Grounded γ Derivation

The message-layer models (analytical, DES, slot-sim) all implement the same sizing equations; their agreement is expected by construction. To anchor the MAC efficiency parameter in physical-layer standards rather than assuming it, we construct a packet-level TDMA simulator that derives γ from CCSDS-standard framing.

1) γ Decomposition from Physical-Layer First Principles: Each TDMA slot is decomposed into four sub-efficiencies (Table XVIII):

$$\gamma_{\text{total}} = \gamma_{\text{framing}} \times \gamma_{\text{FEC}} \times \gamma_{\text{guard}} \times \gamma_{\text{acq}} \tag{13}$$

The derived $\gamma = 0.760$ is below the $\gamma = 0.949$ from the slot-level model (Eq. 4), confirming that FEC and antenna acquisition collectively consume ∼20% of slot capacity.

Rate dependence. γ is weakly rate-dependent through the guard/acquisition term in Eq. 14: at 24 kbps, $\gamma = 0.760$; at 30 kbps, $\gamma = 0.745$ (guard and acquisition consume a larger *bit-equivalent* fraction at higher PHY rates). The 30 kbps feasibility conclusion is robust to this variation: required ingress at $\gamma = 0.745$ is 27.5 kbps $<$ 30 kbps. We report $\gamma = 0.76$ (derived at the 24 kbps threshold) as the conservative reference value; practitioners should evaluate Eq. 14 at their specific PHY rate.

2) Cross-Model Verification: At 30 kbps with $\gamma = 0.745$, all four models (analytical, DES, slot-level, packet-level) agree on ingress ≈ 9,445 ms with ≈ 355 ms margin (Table XX). At 24 kbps, slot-level and packet-level models confirm infeasibility (ingress exceeds T_c). ARQ coupling at 24 kbps produces 52.7% deadline misses (Table XI).

3) FEC Rate Sensitivity: FEC rate has first-order impact: rate-1/2 LDPC makes 30 kbps infeasible; rate-7/8 confirms 30 kbps as minimum viable; rate-1.0 (no FEC) makes 24 kbps feasible at higher BER. Practitioners select FEC appropriate to their link; Eq. 14 accepts the resulting γ.

4) Interpretation: The standards-grounded γ derivation reveals that the feasibility boundary is tighter than a message-layer-only analysis would suggest. At 24 kbps with CCSDS-standard framing (rate-7/8 LDPC, 5 ms acquisition dwell), $\gamma = 0.76$ places the superframe in the infeasible region of Table VI. The 30 kbps design point is confirmed as a physical necessity, not merely a design margin.

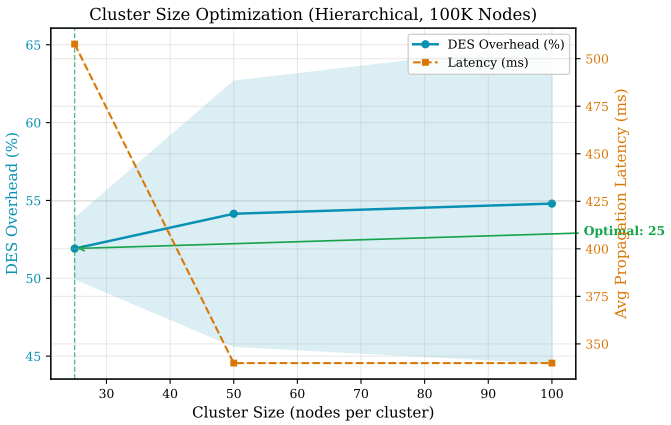

Fig. 14. Cluster size sensitivity: η invariant ($\pm 0.1\%$); latency drives the trade-off. $k_c = 100$–200 favorable.

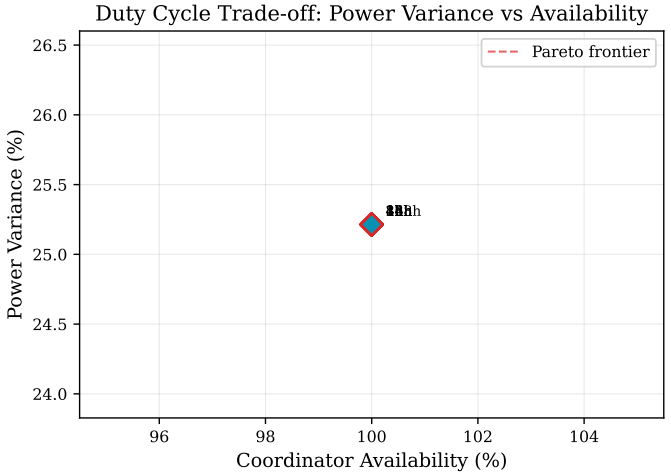

Fig. 15. Duty cycle trade-off in the power variance–system availability space. The 24 h and 48 h configurations offer the best balance.

V. Discussion

A. Validation Gap and Open Questions

Table XIX maps each key result to its verification status. The slot-level TDMA simulator provides model verification (superframe timing, pipeline decoupling, ARQ coupling); the standards-grounded γ derivation (Section IV-I) anchors the MAC efficiency parameter in CCSDS framing rather than assuming it. The DES confirms implementation consistency (expected by construction) and provides distributional analysis of coordinator ingress variability under stochastic campaigns (Fig. 10). Remaining validation—full MAC contention, antenna scheduling, multi-cluster shared-medium effects—requires NS-3 packet-level simulation.

B. Limitations

Physical-layer abstraction. MAC contention, link acquisition, and Earth-occlusion outages are not modeled. **Priority queueing:** equal-priority messages; collision alerts should receive priority. **Failure models:** i.i.d. exponential; correlated modes are future work.

TABLE XVII
Coordination Success vs. Link Availability ($N = 10^4$, $k_c = 100$). $M_r = 2$ columns apply to **Regime A only** ($\geq$10 kbps or $k_c \leq 30$); for RF-backup Regime B (24 kbps, $k_c = 100$), use $M_r = 0$ columns.

p_{link}	Msg. Delivery (%)[b]		Delivered η (%)		Offered[a]
	$M_r=0$	$M_r=2$	$M_r=0$	$M_r=2$	$M_r=2$
1.0	100.0	100.0	46.0	46.0	46.0
0.9	89.9	99.9	41.3	46.0	51.1
0.8	79.9	99.3	36.8	45.7	57.0
0.7	69.8	98.0	32.1	45.1	63.9
0.6	59.7	95.8	27.5	44.1	71.8
0.5	49.9	92.9	23.0	42.7	80.5[b]
0.4	39.9	89.0	18.4	40.9	90.2[b]

[a]Including retransmissions. [b]Per-message rate; per-cycle $\approx (1 - p_{\text{loss}})^{k_c}$.

TABLE XVIII
γ Decomposition from CCSDS Proximity-1 Framing ($S_{\text{EPH}} = 256$ B, 24 kbps PHY)

Component	Mechanism	Efficiency	Source
Framing	ASM+addr+ctrl+FCS: 104 b OH	0.952	CCSDS 211.0-B-5 [
FEC	LDPC rate 7/8: 2152→2460 b	0.875	CCSDS 131.0-B-4 [
Guard	prop+turn+jitter: 4.7 ms	0.956	Prox-1 + geometry
Acquisition	pointing dwell: 5 ms	0.955	Typical S-band ISl
Total derived		**0.760**	Product
Slot-level (Eq. 4)		0.949	No FEC/acq

Dynamic topology. J2 perturbation analysis (Walker constellation: 53°, 550 km, 72 planes × 22 sats) shows co-orbital clusters experience zero re-association; cross-plane events occur at ∼0.014/orbit fleet-wide (∼347-day period per boundary). Re-association overhead: 14.8 kB (Raft election + state handoff), amortized <0.3% of byte budget; AoI transient 33 s (<8% of P99). Cluster membership is fixed in the DES; dynamic membership is future work.

C. Message-Layer Design Equations Summary

For convenience, the key message-layer sizing relationships are collected here. All values assume message-layer operation; multiply bandwidth by $1/\gamma$ for MAC overhead ($\gamma = 0.76$ at 24 kbps, $\gamma = 0.745$ at 30 kbps; standards-derived, Section IV-I; Table XX). Practitioners may compute γ for their specific link from:

$$\gamma = \frac{S \times 8}{\dfrac{S \times 8 + O_{\text{frame}}}{R_{\text{FEC}}} + (T_{\text{guard}} + T_{\text{acq}}) \times R_{\text{PHY}} \times 10^{-3}} \quad (14)$$

where S is payload bytes, O_{frame} is framing overhead (bits), R_{FEC} is FEC code rate, $T_{\text{guard}}/T_{\text{acq}}$ are guard and acquisition times (ms), R_{PHY} is PHY rate (bps), and the 10^{-3} converts ms·bps to bits. Table XX evaluates the feasibility boundary across PHY rates. For CCSDS Proximity-1: $O_{\text{frame}} = 104$ bits, yielding $\gamma = 0.760$ at 24 kbps and $\gamma = 0.745$ at 30 kbps. For CCSDS TC Space Data Link (CLTU encoding): $O_{\text{frame}} = 64$ bits, yielding $\gamma = 0.79$—a 4% improvement shifting the minimum viable PHY rate to ∼28 kbps.

- **Overhead:** $\eta = \eta_0 + d \cdot \eta_{\text{cmd}}$; $\eta_0 \approx 5\%$; $\eta_{\text{cmd}} = p_{\text{cmd}} S_{\text{cmd}} \times 8/(C_{\text{node}} T_c)$; $d \in [0, 1]$ is the campaign duty factor (Table XIII).

TABLE XIX
MODEL VALIDITY & CLAIM MAP

Result	Analyt.	DES	Distrib.	Slot-sim	Pkt-Level	Unmod.
$\eta_0 \approx 5\%$	§IV-E	<0.1%	—	—	—	—
$\eta_S \approx 46\%$	§IV-E	<0.1%	CDF	—	—	—
Coord. 24–30 kbps	Eq. 5	—	CDF	Exact	24 infeas.	—
Margin 623 ms	Tbl. VII	—	—	Exact	—	—
γ sensitivity	Tbl. VI	—	—	Conf.	$\gamma = 0.76$	—
ARQ infeas. (GE)	Markov	—	—	52.7%	Conf.	—
1 kbps design	—	—	—	—	Link budget	—
AoI P99 = 440 s	Eq. 12	Conf.	—	—	—	—
GE P95 rec. 4 cyc.	Markov	Conf.	CDF	—	—	—
$L_{\text{cmd}}(q)$ stagger	Eq. 8	—	—	Conf.	—	—
TDMA decoupling	—	—	—	$(M_r = 0)$	—	—
ARQ coupling	—	—	—	$(M_r \geq 1)$	—	—
T_c^{fleet} reuse	Eq. 6	—	—	—	—	—
Antenna sched.	—	—	—	—	—	Future
MAC contention	—	—	—	—	—	Future

TABLE XX
PHY RATE FEASIBILITY ($k_c = 100$, CCSDS PROX-1 FRAMING)

R_{PHY} (kbps)	$\gamma(R_{\text{PHY}})$	Slot (ms)	Ingress (ms)	Margin (ms)	Feasible?
24	0.760	115.5	11,435	−1,635	N
30	0.745	95.4	9,445	355	Y
50	0.725	60.8	6,019	3,781	Y

Ingress $= (k_c - 1) \times$ Slot; Margin $= T_c -$ Ingress $-$ 200 ms (egress). γ computed from Eq. 14 with rate-dependent guard/acquisition.

- **Coordinator ingress:** $C_{\text{coord}} \geq k_c S_{\text{eph}} \times 8/T_c$ ($\geq$ 20.5 kbps at defaults).
- **AoI P99:** $T_c \lceil \ln(0.01)/\ln(1 - p_{\text{exc}}) \rceil$ (440 s at $p_{\text{exc}} = 0.10$).
- **GE bad-state recovery:** $p_{\text{success}} = 1 - p_B^{M_r+1}$ (27.1%); inter-cycle P95 = 4 cycles at $p_{BG} = 0.50$ (Fig. 8).
- **Safe-mode:** $\gamma_{\min} = \eta_{\text{total}}$; nominal survives Slotted ALOHA ($\gamma = 0.36$); stress requires $\gamma \geq 0.67$.
- **Fleet reuse:** $T_c^{\text{fleet}} = \max(T_c, \lceil f_{\text{RF}} N/(k_c F R) \rceil T_c)$; non-binding at $f_{\text{RF}} \leq 1.2\%$.

Uncertainty at MAC boundary. Physical-layer overheads enter through $1/\gamma$ ($= 1.32$). Second-order interactions (contention $\times$ loss, antenna scheduling $\times$ half-duplex) require NS-3 simulation.

VI. CONCLUSION

This study provides a message-layer sizing framework for hierarchical coordination in autonomous space swarms (10^3–10^5 nodes), decomposing feasibility into two layers: byte budget (η) and physical-layer schedulability (γ, TDMA airtime). The framework separates topology-dependent overhead (η_0) from workload-dependent command traffic (η_{cmd}) and characterizes correlated loss recovery via a Gilbert-Elliott Markov chain parameterized as a sensitivity study across (p_{BG}, p_B).

Under the representative instantiation (Table III): $\eta_0 \approx 5\%$; stress-case $\eta_S \approx 46\%$ at $d = 1$, but realistic $d = 0.01$–0.10 yields $\eta \approx 5$–10%; coordinator ingress ≈27 kbps at standards-derived $\gamma = 0.76$; GE inter-cycle P95 recovery in 4 cycles; AoI P99 = 440 s. Standards-grounded γ derivation (Section IV-I) anchors $\gamma = 0.76$ in CCSDS Proximity-1 framing (rate-7/8 LDPC), confirming 30 kbps as the minimum viable design point. An ON/OFF Markov campaign model (Section IV-E) extends the Bernoulli duty-factor to temporally correlated campaigns, showing heavier-tailed coordinator ingress. The framework's value lies in the complete sizing procedure—from message-layer byte budget through PHY schedulability to coordinator capacity—rather than in any single overhead figure. The remaining NS-3 gap (MAC contention, antenna scheduling) is identified in Table XIX.

DATA AVAILABILITY

Source code, configuration, MC datasets, TDMA slot-level simulator, packet-level TDMA simulator, and link budget calculator are at https://github.com/projectdyson/dyson (tag: `paper-02-v-ce`). Parameters: Table III. Environment: Python 3.12, NumPy 2.x, SciPy 1.x, Matplotlib 3.x. Total MC wall-clock time: ∼90 min on commodity hardware.

ACKNOWLEDGMENT

An AI-assisted ideation exercise (Claude 4.6, Gemini 3 Pro, GPT-5.2; see [25]) motivated aspects of the coordinator architecture but is not validated here.

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